

Sequence

Engineering Mathematics–I (CSE-1001)
BTech 1st Year, IIIT Bhopal

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To the Student

This note explains **sequence** in simple words. It is written for students who are meeting the topic properly for the first time. Every idea is built slowly. First we look at a real situation where the idea is needed, then we look at a picture, then we learn the definition, and only then do we work through examples. You do not need any advanced mathematics. You only need school algebra, inequalities, and the meaning of the modulus sign $|x|$.

Read the sections in order. Everything in this chapter rests on one definition. The ϵ – N definition of a limit in Section 6. So do not rush past it. Keep pen and paper beside you: plot the dots yourself, and try each example on your own before reading its solution.

How to use these notes:

1. Attend the class lecture for each topic.
2. Read the corresponding section of these notes carefully, working through each example yourself.
3. Stop at every *Try it yourself* box and attempt it on the spot.
4. Once you have finished a section, attempt the questions from the [companion Question Bank](#) for self-assessment.

What comes next. This note is the foundation for the companion note on *Series*, which asks what happens when the terms of a sequence are added up. Everything here — the ϵ – N definition, the Monotone Convergence Theorem, the Ratio and Root Tests — is used there from the first page. Finish this note before starting that one.

Good luck, and enjoy the topic.

Notation. Throughout, $\mathbb{N} = \{1, 2, 3, \dots\}$ and $\mathbb{R}$ denotes the set of real numbers.

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1 Why Study Sequences?

A sequence is an ordered list of numbers. That sounds too simple to deserve a chapter. The reason it does is this: *anything that happens in steps produces a sequence*, and almost everything an engineer builds happens in steps.

Four situations, one piece of mathematics

1. Medicine in the bloodstream. A patient takes 100 mg of a drug each morning. Over 24 hours the body eliminates half of whatever is present. Let a_n be the amount in the blood just after the n -th dose. Then

$$a_1 = 100, \quad a_{n+1} = \frac{1}{2}a_n + 100.$$

Does the drug build up without limit (dangerous), or settle at a safe steady level? This is a question about a sequence.

2. A calculator finding $\sqrt{2}$. Your calculator has no “square root” hardware. It starts with a guess $a_1 = 2$ and repeatedly applies

$$a_{n+1} = \frac{1}{2} \left(a_n + \frac{2}{a_n} \right).$$

After four steps it already has 1.414214. Why does this work, and how fast?

3. Digital audio. A microphone samples the air pressure 44,100 times per second. The song stored on your phone is literally a sequence (a_n) of numbers. Every operation applied to it — filtering, compression, noise removal — is an operation on sequences.

4. Training a neural network. Gradient descent produces a sequence of parameter values $w_1, w_2, w_3, \dots$. The entire question of whether “training worked” is the question of whether this sequence converges, and to what.

In each case the practical question — is it safe, is it accurate, did it work — becomes the mathematical question *does this sequence converge, and to what limit?* That is what this chapter answers.

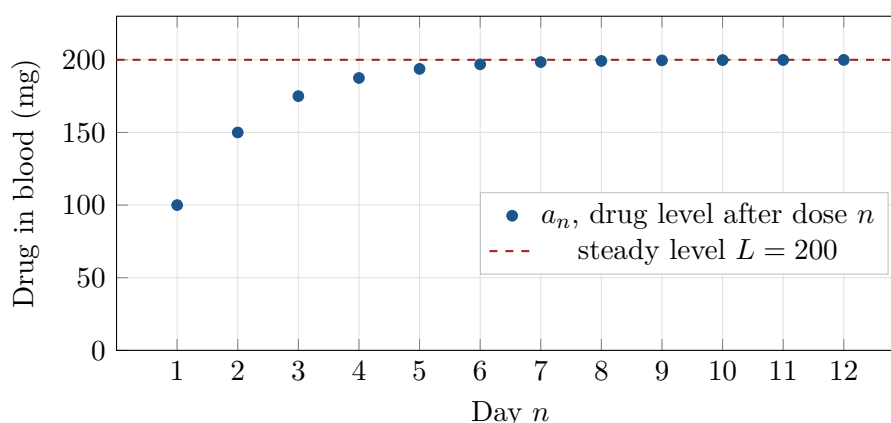


Figure 1: The drug level rises at every dose but never reaches 200 mg. It is *increasing* and *bounded above* — and by the Monotone Convergence Theorem (Section 10) that alone guarantees it settles down. The patient is safe.

The idea

Notice what Figure 1 shows. The terms never actually equal 200. Convergence does not mean “reaching”; it means “getting and staying as close as you like”. Holding on to this distinction is most of the battle.

2 What Is a Sequence?

Definition 2.1 (Sequence). A **sequence** of real numbers is a function $f : \mathbb{N} \rightarrow \mathbb{R}$. Writing a_n for $f(n)$, we denote the sequence by

$$(a_n)_{n=1}^{\infty} \quad \text{or simply} \quad (a_n), \quad \text{listed as} \quad a_1, a_2, a_3, \dots$$

The number a_n is called the **n -th term** or **general term**.

The definition as a *function* is worth pausing on. It says that a sequence assigns exactly one real number to each natural number. The input is the position n ; the output is the term a_n . A sequence is therefore an infinite list with an address attached to every entry.

Example 2.2 (Three ways to describe a sequence). A sequence can be given by:

- (i) **An explicit formula:** $a_n = \frac{1}{n}$, giving $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$
- (ii) **A recursion:** $a_1 = 1, a_2 = 1, a_n = a_{n-1} + a_{n-2}$ for $n \geq 3$, giving the **Fibonacci sequence** $1, 1, 2, 3, 5, 8, 13, 21, \dots$
- (iii) **A description:** $a_n =$ the n -th prime number, giving $2, 3, 5, 7, 11, 13, \dots$ (no formula is known, but the sequence is perfectly well defined).

A sequence is not a set

Order matters in a sequence, and repetition matters.

$$(1, 2, 3, \dots) \neq (3, 2, 1, \dots) \quad \text{but} \quad \{1, 2, 3\} = \{3, 2, 1\}.$$

The sequence $1, 1, 1, \dots$ has infinitely many terms; the set $\{1, 1, 1\} = \{1\}$ has one element. When we write (a_n) with round brackets we mean the ordered list; $\{a_n : n \in \mathbb{N}\}$ with braces means the set of values it takes.

2.1 Two familiar families

Example 2.3 (Arithmetic progression). Each term differs from the previous one by a fixed constant d :

$$2, 5, 8, 11, 14, \dots \quad a_1 = 2, \quad d = 3, \quad a_n = a_1 + (n - 1)d.$$

An AP models anything growing by equal absolute amounts — a fixed monthly saving, a conveyor belt advancing by a fixed distance each step.

Example 2.4 (Geometric progression). Each term is the previous one multiplied by a fixed constant r :

$$3, 6, 12, 24, 48, \dots \quad a_1 = 3, \quad r = 2, \quad a_n = a_1 r^{n-1}.$$

A GP models anything growing or decaying by equal *proportions* — compound interest, radioactive decay, the successive bounce heights of a ball, signal attenuation along a cable.

Which model is which?

A ball dropped from 2 m returns to 60% of its previous height on each bounce. The bounce heights are

$$2, 1.2, 0.72, 0.432, \dots \quad a_n = 2(0.6)^{n-1},$$

a GP with $r = 0.6$. Because $|r| < 1$, the heights tend to 0: the ball stops bouncing — though, interestingly, only after infinitely many bounces in a *finite* total time. The behaviour of r^n for different r is recorded in Section 14.

3 Seeing a Sequence

Before any theory, it helps enormously to know what a sequence *looks like*. There are two standard pictures, and each reveals something different.

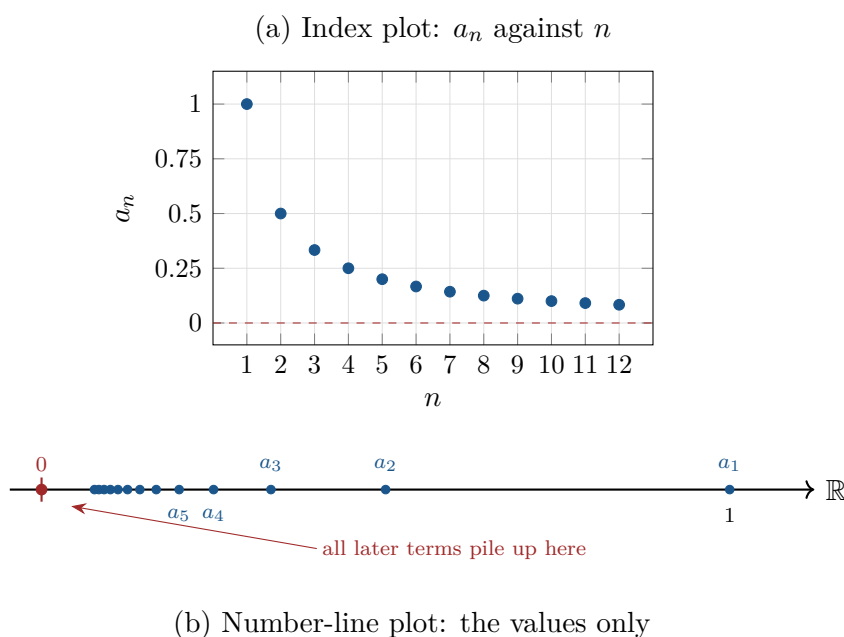


Figure 2: Two views of $a_n = 1/n$. The index plot (a) shows the terms marching steadily downwards towards the dashed line $y = 0$. The number-line plot (b) shows them crowding in on the single point 0. Convergence means exactly this crowding: eventually *all* the terms are inside any small interval around 0 you care to draw.

Four things a sequence can do

Every sequence falls into one of four behaviours, shown in Figure 3. It can settle down to a number (converge); it can run away to $+\infty$ or $-\infty$; it can bounce forever within a fixed range without settling; or it can bounce with ever-growing swings. The whole vocabulary of this chapter — bounded, monotone, convergent, oscillatory — exists to tell these four apart precisely.

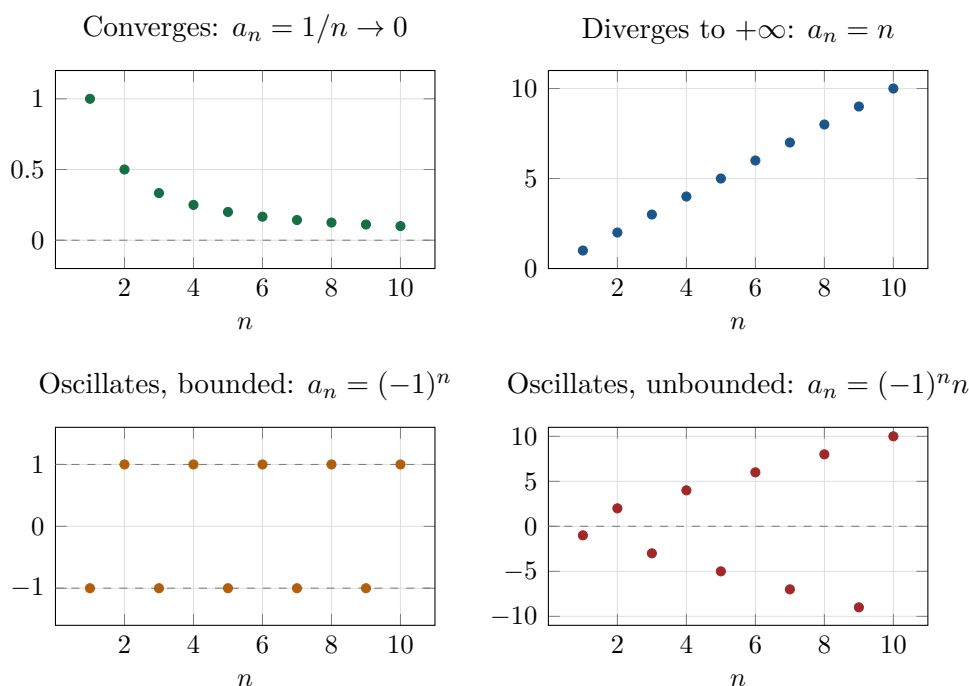


Figure 3: The four possible long-run behaviours of a real sequence. Only the first is called convergent; the other three are all forms of divergence.

Try it yourself

Sketch the first eight terms of each of the following and decide, from the picture alone, which of the four behaviours it shows.

$$(i) a_n = 2 + \frac{(-1)^n}{n} \quad (ii) a_n = \frac{n}{n+1} \quad (iii) a_n = n(-1)^n + n$$

4 Bounded and Unbounded Sequences

The first question to ask about a sequence is whether it stays within a fixed range.

Definition 4.1 (Bounded above, bounded below, bounded). A sequence (a_n) is

- **bounded above** if there is $M \in \mathbb{R}$ with $a_n \leq M$ for all n ; such an M is an **upper bound**;
- **bounded below** if there is $m \in \mathbb{R}$ with $a_n \geq m$ for all n ; such an m is a **lower bound**;

- **bounded** if it is both. Equivalently, there is $K > 0$ with

$$|a_n| \leq K \quad \text{for all } n \in \mathbb{N}.$$

A sequence that is not bounded is **unbounded**: for every $K > 0$ there exists at least one n with $|a_n| > K$.

The idea

“Bounded” means you can draw two horizontal lines on the index plot such that *every* dot lies between them. In Figure 3, the first and third pictures are bounded; the second and fourth are not.

“For all n ” means all of them

A bound must work for every single term, not just from some point onwards, and not just for the terms you checked. Writing “ $a_n \leq 1$ for large n , so the sequence is bounded” is incomplete. (It happens to give the right answer here, because finitely many remaining terms always have a largest one — but that argument must be *stated*, not assumed.)

Example 4.2 (Bounded). (i) $a_n = \frac{1}{n}$: here $0 < a_n \leq 1$, so bounded with lower bound 0 and upper bound 1.

(ii) $a_n = (-1)^n$: $|a_n| = 1$ for all n , so bounded by $K = 1$.

(iii) $a_n = \frac{n}{n+1}$: since $0 < \frac{n}{n+1} < 1$, bounded.

(iv) $a_n = \sin n$: $-1 \leq \sin n \leq 1$, so bounded — yet it is neither increasing nor decreasing. *Bounded does not imply monotone.*

Example 4.3 (Unbounded). (i) $a_n = n$: unbounded above, but bounded below by 1.

(ii) $a_n = -n$: unbounded below, bounded above by -1 .

(iii) $a_n = (-1)^n n$: unbounded above *and* below.

(iv) $a_n = n^2$: unbounded above.

Try it yourself

Is $a_n = \frac{n^2 + 1}{n}$ bounded? Is it bounded below? Find the best (i.e. largest) lower bound you can.

Hint: write it as $n + \frac{1}{n}$ and use AM–GM.

5 Monotone Sequences

The second structural question is whether a sequence moves consistently in one direction.

Definition 5.1 (Monotone sequences). A sequence (a_n) is

- **monotonically increasing** if $a_n \leq a_{n+1}$ for all n ; **strictly increasing** if $a_n < a_{n+1}$ for all n ;
- **monotonically decreasing** if $a_n \geq a_{n+1}$ for all n ; **strictly decreasing** if $a_n > a_{n+1}$ for all n ;
- **monotone** if it is one of increasing or decreasing.

5.1 Three techniques for checking monotonicity

Method	What you compute	When to use it
Difference	Sign of $a_{n+1} - a_n$	Default choice; best for sums, polynomials, rational expressions
Ratio	Compare $\frac{a_{n+1}}{a_n}$ with 1 (terms positive)	Best when a_n involves factorials or powers
Derivative	Sign of $f'(x)$ for $x \geq 1$, where $a_n = f(n)$	Best when f is a smooth formula and the algebra is messy

Example 5.2 (Difference method). For $a_n = \frac{n}{n+1}$,

$$a_{n+1} - a_n = \frac{n+1}{n+2} - \frac{n}{n+1} = \frac{(n+1)^2 - n(n+2)}{(n+1)(n+2)} = \frac{1}{(n+1)(n+2)} > 0,$$

so the sequence is strictly increasing.

Example 5.3 (Ratio method). For $a_n = \frac{2^n}{n!}$, all terms are positive and

$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1}}{(n+1)!} \cdot \frac{n!}{2^n} = \frac{2}{n+1} \leq 1 \quad \text{for } n \geq 1,$$

so $a_{n+1} \leq a_n$: the sequence is monotonically decreasing. (For $n \geq 2$ the inequality is strict; note $a_1 = a_2 = 2$.)

Example 5.4 (Derivative method). For $a_n = \frac{\ln n}{n}$, put $f(x) = \frac{\ln x}{x}$. Then

$$f'(x) = \frac{1 - \ln x}{x^2} < 0 \quad \text{for } x > e,$$

so (a_n) is decreasing from $n = 3$ onwards. It is *eventually* monotone, which is all the Monotone Convergence Theorem ever needs.

Example 5.5 (Not monotone). $a_n = 2 + \frac{(-1)^n}{n}$ gives 1, 2.5, 1.667, 2.25, ..., jumping above and below 2 alternately. It is bounded and (as we shall see) convergent, but not monotone.

Checking the first few terms is not a proof

Students often compute a_1, a_2, a_3 , see them increase, and conclude the sequence is increasing. Consider $a_n = n(7 - n)$: the terms 6, 10, 12, 12, 10, ... increase and then decrease. Monotonicity is a statement about *all* n and needs an argument valid for all n .

6 The Limit of a Sequence

Now we make “settles down to” precise. This is the hardest definition in the first-year syllabus, so we approach it slowly.

The challenge game

Think of convergence as a game between two players.

- **The challenger** names a tolerance $\epsilon > 0$ — as small as they like. Say $\epsilon = 0.001$. This draws a horizontal band of half-width ϵ around the proposed limit L .
- **You** must respond with a cut-off N and guarantee: *every* term from a_N onwards lies inside that band.

If you can meet *every* challenge, however small, then $a_n \rightarrow L$. If there is even one ϵ you cannot answer, the sequence does not converge to L .

Two features of the game matter. First, N is allowed to depend on ϵ : a smaller tolerance may need a longer wait. Second, only finitely many terms are ever allowed to sit outside the band.

Definition 6.1 (Limit of a sequence). A real number L is the **limit** of (a_n) if for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$|a_n - L| < \epsilon \quad \text{for all } n \geq N.$$

We write $\lim_{n \rightarrow \infty} a_n = L$, or $a_n \rightarrow L$.

Recall that $|a_n - L| < \epsilon$ says exactly $L - \epsilon < a_n < L + \epsilon$; that is, a_n lies in the band. Figure 4 draws one round of the game.

6.1 Proving a limit from the definition

The structure of every such proof is the same: *start from ϵ , work backwards to find N , then write the argument forwards.*

Example 6.2. Prove that $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$.

Solution. Scratch work (not part of the final proof). We need $\left| \frac{1}{n} - 0 \right| = \frac{1}{n} < \epsilon$, i.e. $n > \frac{1}{\epsilon}$. So choosing any $N > \frac{1}{\epsilon}$ will do.

Proof. Let $\epsilon > 0$ be given. By the Archimedean property choose $N \in \mathbb{N}$ with $N > \frac{1}{\epsilon}$. Then for all $n \geq N$,

$$\left| \frac{1}{n} - 0 \right| = \frac{1}{n} \leq \frac{1}{N} < \epsilon.$$

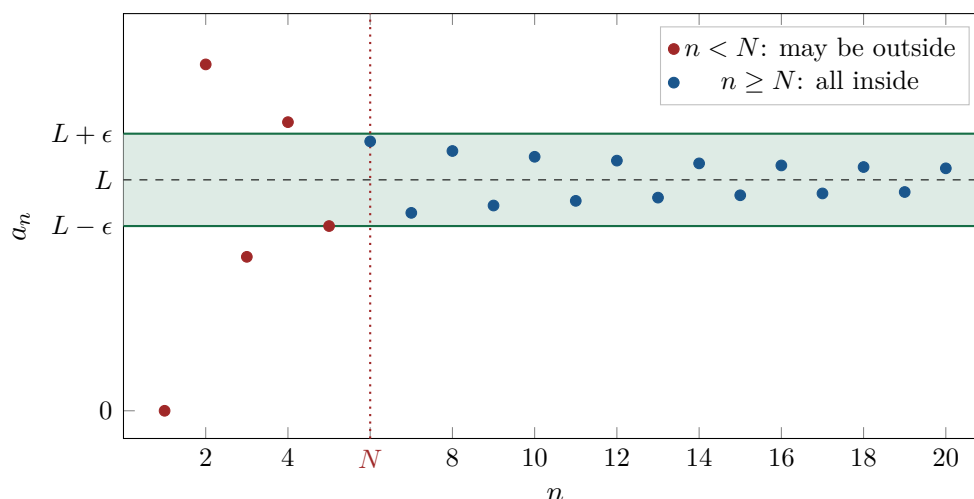


Figure 4: The ϵ - N definition in action for $a_n = 1 + \frac{(-1)^n}{n}$ with $L = 1$ and $\epsilon = 0.2$. The first few terms (red) fall outside the green band — that is permitted, because there are only finitely many of them. From $N = 6$ onwards every term (blue) is inside, and stays inside. Look carefully at $a_5 = 0.8$: it sits exactly *on* the lower edge, and the definition demands the strict inequality $|a_n - L| < \epsilon$, so a_5 does not count as inside and the cut-off is $N = 6$, not 5. A smaller ϵ simply pushes N further right: $\epsilon = 0.01$ needs $N = 101$.

Hence $\frac{1}{n} \rightarrow 0$. □

Example 6.3. Prove that $\lim_{n \rightarrow \infty} \frac{n+1}{n} = 1$.

Solution. Here $\left| \frac{n+1}{n} - 1 \right| = \frac{1}{n}$. Given $\epsilon > 0$, choose $N > \frac{1}{\epsilon}$; then for $n \geq N$ we get $\left| \frac{n+1}{n} - 1 \right| = \frac{1}{n} \leq \frac{1}{N} < \epsilon$. □

Try it yourself

Use the definition to prove that $\lim_{n \rightarrow \infty} \frac{2n}{n+3} = 2$.

Hint: show $\left| \frac{2n}{n+3} - 2 \right| = \frac{6}{n+3} < \frac{6}{n}$, then choose $N > 6/\epsilon$.

The order of the quantifiers

The definition reads: *for every* ϵ , *there exists* N . Never the other way round. A single N that works for one particular ϵ proves nothing; N must be produced *after* ϵ is given, and is normally a formula in ϵ . Writing “take $\epsilon = 0.001$ ” at the start of a proof is the most common error in this topic.

7 Convergent and Divergent Sequences

Definition 7.1 (Convergent, divergent). A sequence (a_n) is **convergent** if there exists a real number L with $\lim_{n \rightarrow \infty} a_n = L$. Otherwise it is **divergent**.

Example 7.2. $\left(\frac{1}{n}\right) \rightarrow 0$; $\left(1 + \frac{1}{n}\right) \rightarrow 1$; the constant sequence $(c, c, c, \dots) \rightarrow c$.

Two basic facts follow immediately from the definition, and both are used constantly afterwards.

Theorem 7.3 (Uniqueness of the limit). *A convergent sequence has exactly one limit.*

Proof. Suppose $a_n \rightarrow L$ and $a_n \rightarrow L'$ with $L \neq L'$. Put $\epsilon = \frac{|L-L'|}{2} > 0$. There are N_1, N_2 with $|a_n - L| < \epsilon$ for $n \geq N_1$ and $|a_n - L'| < \epsilon$ for $n \geq N_2$. For $n \geq \max(N_1, N_2)$, the triangle inequality gives

$$|L - L'| \leq |L - a_n| + |a_n - L'| < \epsilon + \epsilon = |L - L'|,$$

a contradiction. Hence $L = L'$. □

Theorem 7.4 (Convergent $\Rightarrow$ bounded). *Every convergent sequence is bounded.*

Proof. Let $a_n \rightarrow L$. Taking $\epsilon = 1$, there is N with $|a_n - L| < 1$ for all $n \geq N$, hence $|a_n| < |L| + 1$ for those n . Put

$$K = \max\{|a_1|, |a_2|, \dots, |a_{N-1}|, |L| + 1\},$$

a maximum of finitely many numbers. Then $|a_n| \leq K$ for every n . □

How this theorem is actually used

Read it backwards. *An unbounded sequence must diverge.* That is the form you will use in examinations: to show $a_n = n^2/(n+1)$ diverges, it is enough to show it is unbounded. Note also that the converse fails — $(-1)^n$ is bounded but divergent — so boundedness alone is never enough to conclude convergence.

7.1 Types of divergence

Definition 7.5 (Divergence to infinity). $a_n \rightarrow +\infty$ if for every $M > 0$ there is N with $a_n > M$ for all $n \geq N$. Similarly $a_n \rightarrow -\infty$ if for every $M > 0$ there is N with $a_n < -M$ for all $n \geq N$.

A divergent sequence is therefore of one of three kinds:

- (i) divergent to $+\infty$, e.g. $a_n = n$ or $a_n = 2^n$;
- (ii) divergent to $-\infty$, e.g. $a_n = -n$;
- (iii) **oscillatory**, i.e. neither convergent nor tending to $\pm\infty$. This splits further into *bounded* oscillation, such as $a_n = (-1)^n$, and *unbounded* oscillation, such as $a_n = (-1)^{n^2}$.

“Tends to infinity” is still divergence

The phrase $a_n \rightarrow +\infty$ uses the arrow, but $+\infty$ is not a real number and (a_n) is *not* convergent. Writing “ $\lim_{n \rightarrow \infty} n = \infty$, so the sequence converges to ∞ ” is wrong. The notation records a precise kind of divergence.

8 Limit Theorems

Proving every limit from the definition would be exhausting. The next theorem lets us compute most limits algebraically.

Theorem 8.1 (Algebra of limits). *Suppose $\lim_{n \rightarrow \infty} a_n = A$ and $\lim_{n \rightarrow \infty} b_n = B$. Then*

- (i) $\lim(a_n \pm b_n) = A \pm B$;
- (ii) $\lim(ca_n) = cA$ for any constant $c \in \mathbb{R}$;
- (iii) $\lim(a_nb_n) = AB$;
- (iv) if $B \neq 0$ and $b_n \neq 0$ for all n , then $\lim \frac{a_n}{b_n} = \frac{A}{B}$.

The hypotheses are not optional

Every part of Theorem 8.1 assumes that (a_n) and (b_n) *already converge*. It says nothing when they do not. For example, with $a_n = n$ and $b_n = -n$, both diverge, yet $a_n + b_n = 0 \rightarrow 0$. You may not conclude “ $\infty - \infty = 0$ ”; there is simply no theorem being applied.

Example 8.2 (The standard technique for rational expressions). Evaluate $\lim_{n \rightarrow \infty} \frac{3n^2 + 2n + 1}{n^2 + 5}$.

Solution. Divide numerator and denominator by the highest power of n present, here n^2 :

$$\frac{3n^2 + 2n + 1}{n^2 + 5} = \frac{\frac{3n^2}{n^2} + \frac{2n}{n^2} + \frac{1}{n^2}}{\frac{n^2}{n^2} + \frac{5}{n^2}} = \frac{3 + \frac{2}{n} + \frac{1}{n^2}}{1 + \frac{5}{n^2}}.$$

Since $\frac{1}{n} \rightarrow 0$ and $\frac{1}{n^2} \rightarrow 0$, Theorem 8.1 applies to numerator and denominator separately, and the denominator has limit $1 \neq 0$. Hence the limit is $\frac{3 + 0 + 0}{1 + 0} = 3$. $\square$

The idea

The division step is the whole method. Dividing by the dominant power turns every term into either a constant or something of the form $c/n^k \rightarrow 0$. The same trick handles $\frac{5n^3 - 2}{2n^3 + n} \rightarrow \frac{5}{2}$ and $\frac{n + 1}{n^2 + 3} \rightarrow 0$.

9 The Sandwich Theorem

Some sequences are too awkward for the algebra of limits — anything involving $\sin n$ or $\cos n$, for instance, since those have no limit at all. The Sandwich Theorem handles them by trapping the sequence between two others.

Theorem 9.1 (Sandwich / Squeeze Theorem). *Suppose $a_n \leq b_n \leq c_n$ for all $n \geq N_0$, and*

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L.$$

Then (b_n) converges and $\lim_{n \rightarrow \infty} b_n = L$.

The idea

Two walls closing in on the same point leave whatever is trapped between them nowhere to go. Note that the hypothesis is only needed “for all $n \geq N_0$ ” — a reminder that convergence depends on the tail alone.

Example 9.2. Show that $\lim_{n \rightarrow \infty} \frac{\sin n}{n} = 0$.

Solution. For every n we have $-1 \leq \sin n \leq 1$, so dividing by $n > 0$,

$$-\frac{1}{n} \leq \frac{\sin n}{n} \leq \frac{1}{n}.$$

Both outer sequences tend to 0, so by the Sandwich Theorem $\frac{\sin n}{n} \rightarrow 0$. $\square$

Example 9.3. Show that $\lim_{n \rightarrow \infty} \frac{\cos n}{n^2} = 0$.

Solution. Since $|\cos n| \leq 1$, we have $-\frac{1}{n^2} \leq \frac{\cos n}{n^2} \leq \frac{1}{n^2}$, and both bounds tend to 0. $\square$

Example 9.4 (A less obvious sandwich). Show that

$$\lim_{n \rightarrow \infty} \left(\frac{1}{n^2 + 1} + \frac{1}{n^2 + 2} + \cdots + \frac{1}{n^2 + n} \right) = 0.$$

Solution. There are n terms, each between $\frac{1}{n^2 + n}$ and $\frac{1}{n^2 + 1}$. Hence the sum S_n satisfies

$$\frac{n}{n^2 + n} \leq S_n \leq \frac{n}{n^2 + 1}.$$

The left side is $\frac{1}{n+1} \rightarrow 0$ and the right side is $\frac{n}{n^2 + 1} \rightarrow 0$, so $S_n \rightarrow 0$. $\square$

Try it yourself

Use the Sandwich Theorem to show $\lim_{n \rightarrow \infty} \frac{(-1)^n}{\sqrt{n}} = 0$ and $\lim_{n \rightarrow \infty} \frac{n + \sin n}{n} = 1$.

10 Monotone Convergence Theorem

We now reach the most useful theorem in the chapter. Its power is that it proves a limit *exists* without telling us its value — and in engineering practice, knowing that an iteration settles down is often the whole question.

Theorem 10.1 (Monotone Convergence Theorem). (i) If (a_n) is monotonically increasing and bounded above, then (a_n) converges, and its limit is $\sup\{a_n : n \in \mathbb{N}\}$, the least upper bound.

(ii) If (a_n) is monotonically decreasing and bounded below, then (a_n) converges, and its limit is $\inf\{a_n : n \in \mathbb{N}\}$, the greatest lower bound.

In short: every bounded monotone sequence converges.

Theorem 10.2 (Companion statement). A monotonically increasing sequence that is not bounded above diverges to $+\infty$; a monotonically decreasing sequence not bounded below diverges to $-\infty$.

Monotone sequences cannot oscillate

Putting the two theorems together: a monotone sequence either converges or runs off to $\pm\infty$. The messy third possibility — oscillation — is impossible for it. That is why monotonicity is so valuable: it reduces four behaviours to two, and boundedness then picks between them.

Example 10.3. $a_n = 1 - \frac{1}{n}$ is strictly increasing and bounded above by 1, hence convergent; indeed $a_n \rightarrow 1$. Note that 1 is never attained: the limit is the least upper bound, not a term of the sequence.

Example 10.4. $a_n = \frac{1}{n}$ is strictly decreasing and bounded below by 0, hence convergent, with limit $\inf\{1/n\} = 0$.

10.1 Worked application: the drug dosage model

Finishing the problem from Section 1

Recall $a_1 = 100$ and $a_{n+1} = \frac{1}{2}a_n + 100$.

Step 1: bounded above by 200. Use induction. Certainly $a_1 = 100 < 200$. If $a_n < 200$ then

$$a_{n+1} = \frac{1}{2}a_n + 100 < \frac{1}{2}(200) + 100 = 200.$$

So $a_n < 200$ for all n .

Step 2: increasing. We have

$$a_{n+1} - a_n = \frac{1}{2}a_n + 100 - a_n = 100 - \frac{1}{2}a_n = \frac{1}{2}(200 - a_n) > 0$$

by Step 1. So (a_n) is strictly increasing.

Step 3: conclude and find L . By Theorem 10.1 the sequence converges; call the limit L . Since $a_{n+1} \rightarrow L$ as well, taking limits in the recursion gives

$$L = \frac{1}{2}L + 100 \implies L = 200.$$

Interpretation. The drug level approaches a steady state of 200 mg and never exceeds it. If the toxic threshold is, say, 250 mg, this dosing schedule is safe indefinitely. Figure 1 is exactly this argument in picture form.

Find the limit only after proving it exists

Step 3 assumed $a_n \rightarrow L$ before writing $L = \frac{1}{2}L + 100$. That assumption *must* be justified first. Applying the same manipulation to $a_{n+1} = 2a_n$ with $a_1 = 1$ gives $L = 2L$, hence $L = 0$ — plainly false, since $a_n = 2^{n-1} \rightarrow \infty$. The algebra is only valid once convergence is established.

10.2 Worked application: the Babylonian square-root algorithm

How a calculator finds $\sqrt{2}$

Define $a_1 = 2$ and $a_{n+1} = \frac{1}{2} \left(a_n + \frac{2}{a_n} \right)$.

Bounded below by $\sqrt{2}$. By AM–GM, for $a_n > 0$,

$$a_{n+1} = \frac{a_n + \frac{2}{a_n}}{2} \geq \sqrt{a_n \cdot \frac{2}{a_n}} = \sqrt{2}.$$

Decreasing. For $n \geq 2$ we have $a_n \geq \sqrt{2}$, so $a_n^2 \geq 2$ and

$$a_n - a_{n+1} = a_n - \frac{1}{2} \left(a_n + \frac{2}{a_n} \right) = \frac{a_n^2 - 2}{2a_n} \geq 0.$$

By Theorem 10.1 the sequence converges to some $L \geq \sqrt{2}$, and taking limits in the recursion gives $L = \frac{1}{2}(L + \frac{2}{L})$, i.e. $L^2 = 2$, so $L = \sqrt{2}$.

n	a_n	correct digits
1	2.000000000	0
2	1.500000000	1
3	1.416666667	2
4	1.414215686	5
5	1.414213562	11

The number of correct digits roughly doubles each step — this is *quadratic convergence*, and it is why the method, known to Babylonian mathematicians around 1700 BCE, is still what runs inside modern numerical libraries.

Try it yourself

Let $a_1 = 1$ and $a_{n+1} = \sqrt{2 + a_n}$. Show that (a_n) is increasing and bounded above by 2, and find its limit.

Answer: $L = 2$.

11 Subsequences

Definition 11.1 (Subsequence). Let (a_n) be a sequence and let $n_1 < n_2 < n_3 < \dots$ be a strictly increasing sequence of natural numbers. Then

$$(a_{n_k})_{k=1}^{\infty} = a_{n_1}, a_{n_2}, a_{n_3}, \dots$$

is called a **subsequence** of (a_n) .

A subsequence is what remains after deleting some terms (possibly infinitely many) while keeping the survivors in their original order. The condition $n_1 < n_2 < \dots$ is

essential: you may skip terms but never reorder them, and you must keep infinitely many.

In the real world

A temperature sensor records a reading every second. Taking only the readings at the top of each minute gives a subsequence. If the true temperature is settling towards a value, so must every such sampling — which is Theorem 11.4 below, and the reason sampling a converging process less often still shows convergence.

Example 11.2. For $(a_n) = (1/n)$:

- taking $n_k = 2k$ gives $\frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \dots \rightarrow 0$;
- taking $n_k = k^2$ gives $1, \frac{1}{4}, \frac{1}{9}, \frac{1}{16}, \dots \rightarrow 0$.

Both subsequences converge to the same limit as the parent sequence.

Example 11.3 (The key example). For $a_n = (-1)^n$:

- even indices $n_k = 2k$ give $1, 1, 1, \dots \rightarrow 1$;
- odd indices $n_k = 2k - 1$ give $-1, -1, -1, \dots \rightarrow -1$.

Two subsequences with different limits. The parent sequence therefore diverges — although it is bounded and has plenty of convergent subsequences.

Theorem 11.4. *If $a_n \rightarrow L$, then every subsequence of (a_n) also converges to L .*

Theorem 11.5 (Divergence test via subsequences). *If (a_n) has two subsequences converging to different limits, then (a_n) diverges.*

The practical use

Theorem 11.5 is the cleanest tool for *proving divergence*. To show $a_n = \cos(n\pi)$ diverges, exhibit two subsequences with different limits and stop. No ϵ - N argument is needed.

Theorem 11.6 (Bolzano–Weierstrass). *Every bounded sequence of real numbers has a convergent subsequence.*

The idea

Infinitely many terms crammed into a finite interval must pile up somewhere. The theorem does not say the sequence converges — Example 11.3 shows it may not — only that some sub-collection of its terms does. This is one of the foundational results of analysis and underlies existence proofs throughout optimisation and differential equations.

Try it yourself

Show that $a_n = \sin\left(\frac{n\pi}{2}\right)$ diverges by finding two subsequences with different limits.

Hint: try $n_k = 4k$ and $n_k = 4k + 1$.

12 Cauchy Sequences

Every definition so far that certifies convergence requires knowing the limit L in advance. That is a serious practical problem: when an algorithm is running, the answer is precisely what we do not have. Cauchy's idea removes the limit from the definition entirely.

Definition 12.1 (Cauchy sequence). A sequence (a_n) is a **Cauchy sequence** if for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$|a_m - a_n| < \epsilon \quad \text{for all } m, n \geq N.$$

Limit-free convergence

Compare the two definitions:

Convergent	terms get close <i>to</i> L
Cauchy	terms get close <i>to each other</i>

The Cauchy condition mentions no limit. So it can be checked by inspecting the sequence alone — which is exactly what a program does when it stops iterating once successive answers agree to six decimal places.

Example 12.2. $a_n = \frac{1}{n}$ is Cauchy.

Proof. Let $\epsilon > 0$ and choose $N > \frac{2}{\epsilon}$. For $m, n \geq N$,

$$\left| \frac{1}{m} - \frac{1}{n} \right| \leq \frac{1}{m} + \frac{1}{n} \leq \frac{2}{N} < \epsilon.$$

□

Example 12.3. Every constant sequence $(c, c, c, \dots)$ is Cauchy, since $|a_m - a_n| = 0$ for all m, n .

Example 12.4 (Not Cauchy). $a_n = n$ is not Cauchy.

Proof. Take $\epsilon = \frac{1}{2}$. For any N , choose $m = N + 1$ and $n = N$. Then

$$|a_m - a_n| = 1 > \frac{1}{2} = \epsilon,$$

so no N can work.

□

Example 12.5 (Not Cauchy). $a_n = (-1)^n$ is not Cauchy: consecutive terms always differ by 2, so no $\epsilon < 2$ can be met.

Theorem 12.6 (Cauchy convergence criterion). *A sequence of real numbers converges if and only if it is Cauchy.*

Corollary 12.7. *Every convergent sequence is Cauchy.*

Corollary 12.8. *Every Cauchy sequence is bounded.*

Property	Implication
Convergent	$\implies$ Cauchy
Cauchy	$\implies$ Convergent (in $\mathbb{R}$)
Cauchy	$\implies$ Bounded
Bounded	$\not\implies$ Cauchy (e.g. $(-1)^n$)

Why “in $\mathbb{R}$ ” is not a formality

The criterion is a property of the real numbers called **completeness**, and it fails in $\mathbb{Q}$. The Babylonian sequence 2, 1.5, $1.41\bar{6}$, $1.4142156\dots$ consists entirely of rational numbers and is Cauchy, yet its limit $\sqrt{2}$ is irrational. Inside $\mathbb{Q}$ that sequence bunches up around a hole. Filling in those holes is what constructing $\mathbb{R}$ means, and it is why calculus is done over $\mathbb{R}$ and not $\mathbb{Q}$.

Example 12.9 (A sequence whose terms stop moving but which still diverges). Let

$$a_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} \quad (\text{the harmonic sums}).$$

Consecutive terms satisfy $a_{n+1} - a_n = \frac{1}{n+1} \rightarrow 0$, which tempts students to call it Cauchy. It is not. Taking $m = 2n$,

$$a_{2n} - a_n = \frac{1}{n+1} + \cdots + \frac{1}{2n} \geq n \cdot \frac{1}{2n} = \frac{1}{2},$$

so $|a_m - a_n| \geq \frac{1}{2}$ for infinitely many pairs. Hence (a_n) diverges — in fact $a_n \rightarrow \infty$, though extremely slowly ($a_n > 10$ requires $n > 12,000$).

Consecutive terms are not enough

The Cauchy condition requires $|a_m - a_n| < \epsilon$ for *all* $m, n \geq N$, not merely for $m = n+1$. The harmonic sums above are the standard counterexample: neighbouring terms become arbitrarily close, yet distant terms stay far apart. Checking only $|a_{n+1} - a_n| \rightarrow 0$ proves nothing.

13 Ratio and Root Tests

These two tests answer one specific and very common question quickly: *does $a_n \rightarrow 0$?* They apply to sequences of positive terms, and they are at their best when factorials and powers are involved.

Theorem 13.1 (Ratio test for sequences). *Let $a_n > 0$ and suppose $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = L$ exists. Then*

- (i) $L < 1 \implies a_n \rightarrow 0$;
- (ii) $L > 1 \implies a_n \rightarrow +\infty$ (so (a_n) diverges);
- (iii) $L = 1 \implies$ the test gives no information.

The idea

The ratio a_{n+1}/a_n measures the factor by which the sequence is multiplied at each step. If that factor settles below 1, the sequence is eventually shrinking geometrically, and geometric decay always wins. If it settles above 1, the terms grow geometrically. At exactly 1 the sequence is being multiplied by something tending to 1, and anything can happen — hence the inconclusive case.

Example 13.2. Test $a_n = \frac{2^n}{n!}$ for convergence.

Solution.

$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1}}{(n+1)!} \cdot \frac{n!}{2^n} = \frac{2}{n+1} \rightarrow 0 < 1.$$

By the Ratio Test, $\frac{2^n}{n!} \rightarrow 0$. (Factorials beat exponentials.) □

Example 13.3. Test $a_n = \frac{n!}{n^n}$.

Solution.

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} = \frac{(n+1)n^n}{(n+1)^{n+1}} = \left(\frac{n}{n+1}\right)^n = \frac{1}{\left(1 + \frac{1}{n}\right)^n} \rightarrow \frac{1}{e} < 1.$$

Hence $\frac{n!}{n^n} \rightarrow 0$. (And n^n beats $n!$.) □

Example 13.4 (Inconclusive). For $a_n = \frac{1}{n}$ we get $\frac{a_{n+1}}{a_n} = \frac{n}{n+1} \rightarrow 1$, so the test says nothing — even though $a_n \rightarrow 0$. For $a_n = n$ we also get $L = 1$, and there the sequence diverges. Both outcomes occur at $L = 1$, which is precisely why the case is inconclusive.

Theorem 13.5 (Root test for sequences). *Let $a_n \geq 0$ and suppose $\lim_{n \rightarrow \infty} (a_n)^{1/n} = L$ exists. Then*

(i) $L < 1 \implies a_n \rightarrow 0$;

(ii) $L > 1 \implies (a_n)$ diverges;

(iii) $L = 1 \implies$ the test gives no information.

Example 13.6. Test $a_n = \frac{n}{2^n}$.

Solution.

$$(a_n)^{1/n} = \frac{n^{1/n}}{2} \rightarrow \frac{1}{2} < 1,$$

using the standard limit $n^{1/n} \rightarrow 1$. Hence $\frac{n}{2^n} \rightarrow 0$. □

Example 13.7. Test $a_n = \left(\frac{3n+1}{4n+2}\right)^n$.

Solution. The n -th power makes the root test the natural choice:

$$(a_n)^{1/n} = \frac{3n+1}{4n+2} = \frac{3 + \frac{1}{n}}{4 + \frac{2}{n}} \rightarrow \frac{3}{4} < 1,$$

so $a_n \rightarrow 0$. □

Example 13.8. Test $a_n = \frac{1}{n^n}$. Here $(a_n)^{1/n} = \frac{1}{n} \rightarrow 0 < 1$, so $a_n \rightarrow 0$.

13.1 Choosing between the two tests

Sequence	Ratio test	Root test	Conclusion
$1/n$	$L = 1$ (fails)	$L = 1$ (fails)	$\rightarrow 0$ (by other means)
$2^n/n!$	$L = 0$	$L = 0$	$\rightarrow 0$
$n!/n^n$	$L = 1/e$	$L = 1/e$	$\rightarrow 0$
$n/2^n$	$L = 1/2$	$L = 1/2$	$\rightarrow 0$
2^n	$L = 2$	$L = 2$	diverges

In practice: **use the ratio test when factorials are present**, since they cancel neatly; **use the root test when the whole expression is an n -th power**, since the root cancels it. The next theorem says that when both give an answer, they give the same one.

Theorem 13.9 (Relation between the two tests). *Let $a_n > 0$. If $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = L$ exists, then $\lim_{n \rightarrow \infty} (a_n)^{1/n}$ also exists and equals L .*

Remark 13.10. The converse is false. For $a_n = 2^{-n+(-1)^n}$ the ratios alternate between 2 and $\frac{1}{8}$ and have no limit, while $(a_n)^{1/n} \rightarrow \frac{1}{2}$. So the root test succeeds on sequences where the ratio test does not even apply: *the root test is strictly stronger*. The ratio test remains more popular only because it is usually easier to compute.

14 Standard Limits and the Growth Hierarchy

14.1 Limits worth memorising

Limit	Condition
$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$	$p > 0$
$\lim_{n \rightarrow \infty} r^n = \begin{cases} 0, & r < 1 \\ 1, & r = 1 \\ \text{diverges,} & r > 1 \text{ or } r = -1 \end{cases}$	
$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$	
$\lim_{n \rightarrow \infty} \left(1 + \frac{a}{n}\right)^n = e^a$	$a \in \mathbb{R}$
$\lim_{n \rightarrow \infty} a^{1/n} = 1$	$a > 0$
$\lim_{n \rightarrow \infty} n^{1/n} = 1$	
$\lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0$	
$\lim_{n \rightarrow \infty} n r^n = 0$	$ r < 1$
$\lim_{n \rightarrow \infty} \frac{a^n}{n!} = 0$	$a \in \mathbb{R}$
$\lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0$	
$\lim_{n \rightarrow \infty} \frac{\sin n}{n} = \lim_{n \rightarrow \infty} \frac{\cos n}{n} = 0$	

14.2 The growth hierarchy

Most of the table above is a consequence of one organising principle. For large n ,

$$\ln n \prec n^p \prec a^n \prec n! \prec n^n \quad (p > 0, a > 1)$$

where $f \prec g$ means $\frac{f(n)}{g(n)} \rightarrow 0$: whatever is on the left is eventually negligible compared with whatever is on the right.

Reading a limit straight off the hierarchy

Any quotient with one hierarchy member on top and a later one below tends to 0; reversed, it tends to ∞ . So

$$\frac{\ln n}{n} \rightarrow 0, \quad \frac{n^{100}}{2^n} \rightarrow 0, \quad \frac{2^n}{n!} \rightarrow 0, \quad \frac{n!}{n^n} \rightarrow 0, \quad \frac{n!}{2^n} \rightarrow \infty.$$

This single picture replaces most of the table.

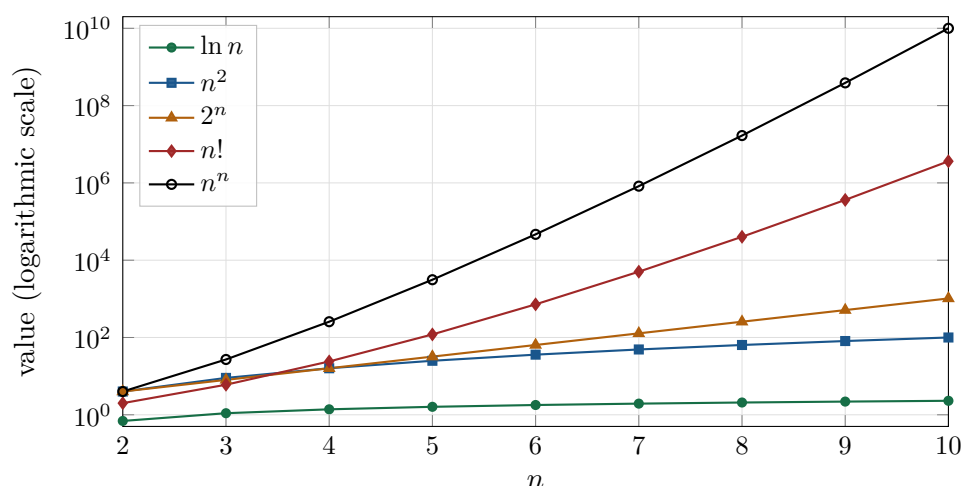


Figure 5: The growth hierarchy on a logarithmic vertical scale. Note the crossovers: n^2 stays ahead of 2^n up to $n = 4$, and 2^n stays ahead of $n!$ up to $n = 3$ — yet each is eventually overtaken and then left far behind. This is the point of the phrase “for large n ”.

Why computer scientists care

Figure 5 is the algorithm complexity chart. An $O(n \log n)$ sorting algorithm and an $O(n^2)$ one may be indistinguishable on 100 items, but on 10^6 items the first finishes in seconds and the second in hours. An $O(2^n)$ or $O(n!)$ algorithm — brute-force travelling salesman, for instance — becomes unusable past $n \approx 20$, whatever hardware you buy. Asymptotic growth is a statement about sequences, and it is decided by exactly the limits in this section.

15 Summary

The logical structure of the chapter is shown in Figure 6; the statements are collected below it.

- A **sequence** is a function $f : \mathbb{N} \rightarrow \mathbb{R}$, written (a_n) . Order matters; it is not a set.
- (a_n) is **bounded** if all terms lie between two fixed numbers; **monotone** if it never changes direction.
- $a_n \rightarrow L$ means: for every $\epsilon > 0$ there is N such that all terms from a_N onwards lie within ϵ of L . Equivalently, all but finitely many terms lie in every band around L .
- A convergent sequence has a **unique** limit and is **bounded**. The converse fails.
- Divergence takes three forms: to $+\infty$, to $-\infty$, or oscillatory.
- The **algebra of limits** and the **Sandwich Theorem** are the two workhorses for computing limits; the sandwich handles $\sin n$ and $\cos n$.
- **Monotone Convergence Theorem**: every bounded monotone sequence converges. This proves existence without producing the value, and is the right tool for recursively defined sequences.

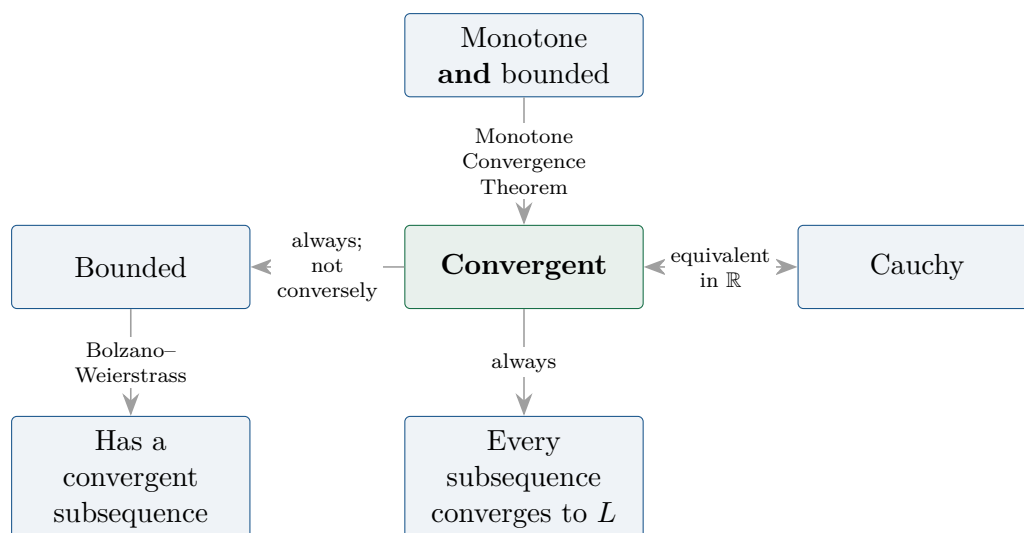


Figure 6: How the main results fit together. Convergence sits at the centre: monotone + bounded is a sufficient condition for it, Cauchy is equivalent to it, and boundedness is a necessary consequence of it.

- A **subsequence** keeps terms in order. If $a_n \rightarrow L$ then every subsequence $\rightarrow L$; two subsequences with different limits prove divergence. Every bounded sequence has a convergent subsequence (**Bolzano–Weierstrass**).
- (a_n) is **Cauchy** if its terms become arbitrarily close to one another. In $\mathbb{R}$ (but not in $\mathbb{Q}$), Cauchy and convergent mean the same thing.
- The **Ratio** and **Root** tests decide whether $a_n \rightarrow 0$ for positive sequences; both are inconclusive at $L = 1$, and the root test is the stronger of the two.
- The **growth hierarchy** $\ln n \prec n^p \prec a^n \prec n! \prec n^n$ recovers most of the standard limits at a glance.

Note: After completing these notes, proceed to the [companion Question Bank](#) for self-assessment.

Best wishes.
